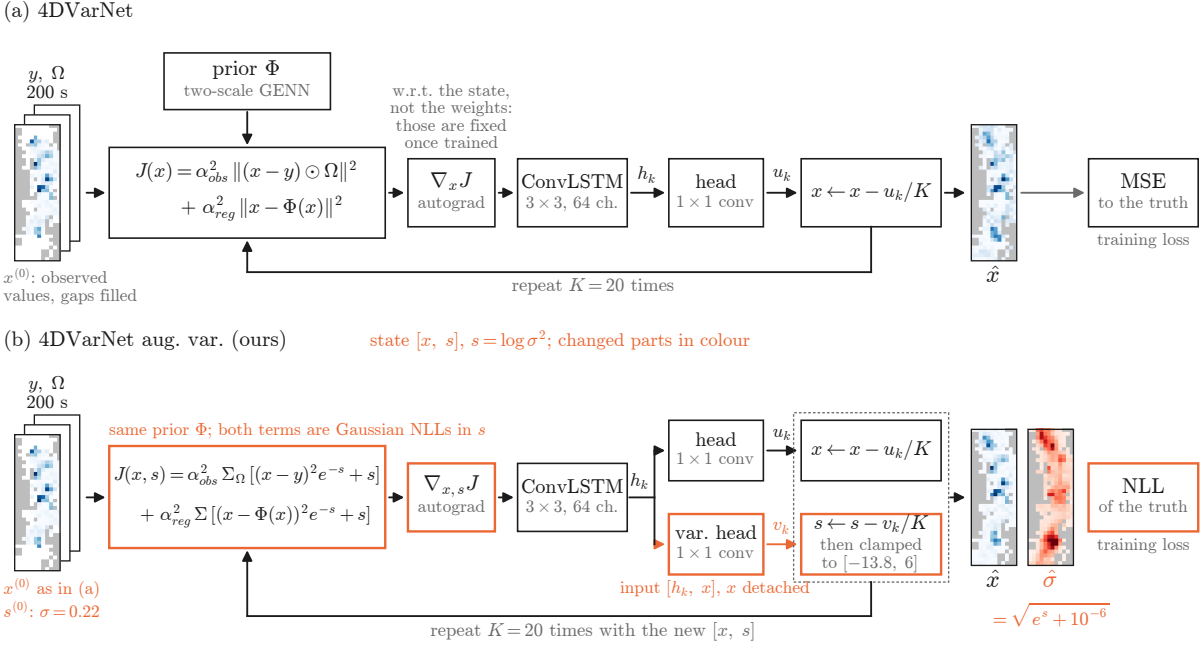
(a) 4DVarNet
y, Ω
200 s
prior Φ
two-scale GENN
w.r.t. the state, not the weights: those are fixed once trained
$J(x) = \alpha_{obs}^2 \|(x-y)\odot\Omega\|^2 + \alpha_{reg}^2 \|x-\Phi(x)\|^2$
$\nabla_x J$ autograd
ConvLSTM 3 × 3, 64 ch.
h_k
head 1 × 1 conv
u_k
$x \leftarrow x - u_k/K$
$\hat{x}$
MSE to the truth
training loss
$x^{(0)}$: observed values, gaps filled
repeat $K = 20$ times
(b) 4DVarNet aug. var. (ours)
state $[x,\ s]$, $s = \log\sigma^2$; changed parts in colour
y, Ω
200 s
same prior Φ; both terms are Gaussian NLLs in s
$J(x,s) = \alpha_{obs}^2 \Sigma_\Omega [(x-y)^2 e^{-s} + s] + \alpha_{reg}^2 \Sigma [(x-\Phi(x))^2 e^{-s} + s]$
$\nabla_{x,s} J$ autograd
ConvLSTM 3 × 3, 64 ch.
h_k
head 1 × 1 conv
u_k
$x \leftarrow x - u_k/K$
var. head 1 × 1 conv
v_k
$s \leftarrow s - v_k/K$ then clamped to $[-13.8,\ 6]$
input $[h_k,\ x]$, x detached
$\hat{x}$
$\hat{\sigma}$
$= \sqrt{e^s + 10^{-6}}$
NLL of the truth
training loss
$x^{(0)}$ as in (a)
$\dot{s}^{(0)}$: $\sigma = 0.22$
repeat $K = 20$ times with the new $[x,\ s]$